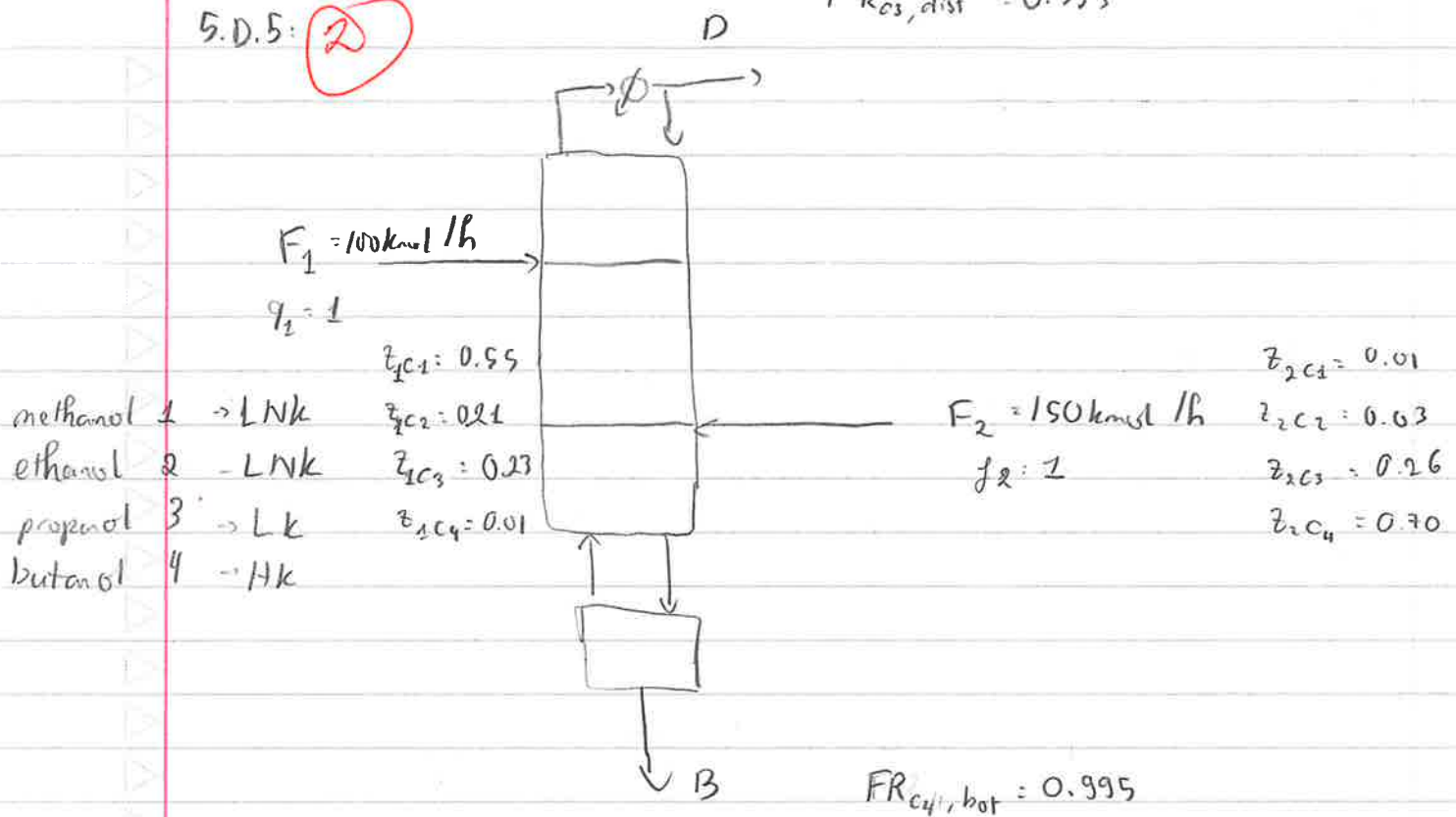


8.5

5.D.5: 2

$$FR_{C3, dist} = 0.993$$



- Assume all LNK evaporated thru the distillate stream
- Assume CMU Condition

Without assumption, we have 10 unknowns (mol frac of each component in B&D, B&D flow rate) and only 8 independent eqns (4 component mass balance, 2 fractional recovery and 2 stack eqns of mol frac of distillate and bot streams), therefore, 2 assumptions on LNK would result in 8 eqns, 8 unknowns, making the system solvable.

$FR_{C3, dist} = 0.993$, it is reasonable to assume $FR_{C1, dist} = FR_{C2, dist} = 1$.

$$FR_{C1, dist} = 1 = \frac{D \cdot x_{C1, dist}}{F_1 \cdot z_{1C1} + F_2 \cdot z_{2C1}} \Rightarrow D \cdot x_{C1, dist} = 100 \cdot 0.55 + 150 \cdot 0.01 = 56.5 \frac{\text{kmol}}{\text{h}}$$

$$\Rightarrow B \cdot x_{C1, bot} = 0 \frac{\text{kmol}}{\text{h}}$$

$$FR_{C2, dist} = 1 = \frac{B \cdot x_{C2, dist}}{F_1 \cdot z_{1C2} + F_2 \cdot z_{2C2}} = B \cdot x_{C2, dist} = 100 \cdot 0.21 + 150 \cdot 0.03 = 25.5 \frac{\text{kmol}}{\text{h}}$$

$$\Rightarrow B \cdot x_{C2, bot} = 0 \frac{\text{kmol}}{\text{h}}$$

$$FR_{C_3 \text{ dist}} = 0.993 = \frac{D x_{C_3 \text{ dist}}}{F_1 z_{1C_3} + F_2 z_{2C_3}} \Rightarrow D x_{C_3 \text{ dist}} = 0.993 (100 \cdot 0.23 + 150 \cdot 0.26)$$

$$D x_{C_3 \text{ dist}} = 61.566 \text{ kmol/h}$$

$$\Rightarrow FR_{C_3 \text{ bot}} = 0.007 = \frac{B x_{C_3 \text{ bot}}}{F_1 z_{1C_3} + F_2 z_{2C_3}} \Rightarrow B x_{C_3 \text{ bot}} = 0.007 (100 \cdot 0.23 + 150 \cdot 0.26)$$

$$B x_{C_3 \text{ bot}} = 0.434 \text{ kmol/h}$$

$$FR_{C_4 \text{ bot}} = 0.995 = \frac{B x_{C_4 \text{ bot}}}{F_1 z_{1C_4} + F_2 z_{2C_4}} \Rightarrow B x_{C_4 \text{ bot}} = 0.995 (100 \cdot 0.01 + 150 \cdot 0.70) = 105.47 \text{ kmol/h}$$

$$FR_{C_4 \text{ top}} = 0.005 = \frac{D x_{C_4 \text{ top}}}{F_1 z_{1C_4} + F_2 z_{2C_4}} \Rightarrow D x_{C_4 \text{ top}} = 0.005 (150 \cdot 0.70 + 100 \cdot 0.01) = 0.53 \text{ kmol/h}$$

$$D = \sum_{i=1}^4 D x_{C_i \text{ top}} = 56.5 + 25.5 + 61.566 + 0.53 = 144.096 \text{ kmol/h}$$

$$B = \sum_{i=1}^4 B x_{C_i \text{ bot}} = 0 + 0 + 0.434 + 105.47 = 105.904 \text{ kmol/h}$$

$$x_{C_1 \text{ dist}} = \frac{D x_{C_1 \text{ dist}}}{D} = \frac{56.5}{144.096} = 0.392$$

$$x_{C_2 \text{ dist}} = \frac{D x_{C_2 \text{ dist}}}{D} = \frac{25.5}{144.096} = 0.177$$

$$x_{C_3 \text{ dist}} = \frac{D x_{C_3 \text{ dist}}}{D} = \frac{61.566}{144.096} = 0.425$$

$$x_{C_4 \text{ dist}} = \frac{0.53}{144.096} = 0.004$$

$$x_{C_1 \text{ bot}} = \frac{B x_{C_1 \text{ bot}}}{B} = 0$$

$$x_{C_2 \text{ bot}} = \frac{B x_{C_2 \text{ bot}}}{B} = 0$$

$$x_{C_3 \text{ bot}} = \frac{B x_{C_3 \text{ bot}}}{B} = \frac{0.434}{105.904} = 0.004$$

$$x_{C_4 \text{ bot}} = \frac{B x_{C_4 \text{ bot}}}{B} = \frac{105.47}{105.904} = 0.996$$

F.D.2

1.2

partial
condenser



$$FR_{1D} = 0.999$$

propylene 1, LK
propane 2, HK
isobutane 3, HNK

$$FR_{2B} \geq 0.950$$

$$FR_{3B} \geq 0.9999$$

Since isobutane fractional recovery is 0.9999 at the bottom, it is basically 1, it would be more appropriate to treat this HNK and propane as HK.

$\Rightarrow$ propylene $\rightarrow$ LK, propane $\rightarrow$ HK, isobutane HNK

$T_{top} = 34^\circ\text{C}$
 $T_{bot} = 40^\circ\text{C}$

Assume the distillation column operates so that $T_{dist} = 34^\circ\text{C}$ and $T_{bot} = 40^\circ\text{C}$

Use Fenske eqn for multi component

$$\frac{FR_{LK\ dist}}{FR_{HK\ dist}} = \alpha_{LK-HK}^{N_{min}} \frac{FR_{LK\ bot}}{FR_{HK\ bot}}$$

$$\alpha_{LK-HK}^{N_{min}} = (\alpha_{1-2\ dist} \alpha_{1-2\ bot})^{N_{min}}$$

$$\Rightarrow \frac{0.999}{1-0.95} = \left(\frac{1.00}{0.85} \frac{1.13}{1.00} \right)^{N_{min}} \frac{1-0.999}{0.95}$$

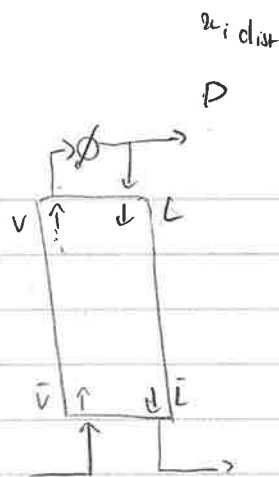
$$\Rightarrow N_{min} = 41.26 = 42 \text{ stages}$$

Since we don't have info of the middle stages

7. D. 9

(2)

benzene 1k
toluene 14k
xylene 401k



$$L/D = 9$$

$$x_{tol, dist} = 0.007$$

$$F = 150 \text{ kmol/h}$$

$$p = 1$$

$$z_{ben} = 0.92$$

$$z_{tol} = 0.385$$

$$z_{xy} = 0.095$$

$$B$$

$$\alpha_{ben-tol} = 2.22$$

$$\alpha_{tol-xy} = 2.01$$

a) CMU condition $\Rightarrow$ $F = \bar{V} = V = 150 \text{ kmol/h}$
 $L = \bar{L} = B$

$$\frac{L}{D} = 9 \Rightarrow \frac{L}{V} = \frac{9}{10} \Rightarrow L = \frac{9}{10} V \Rightarrow B = \frac{9}{10} F = 135 \text{ kmol/h}$$

$$F = B + D \Rightarrow D = F - B = 15 \text{ kmol/h}$$

$$F z_{tol} = D x_{tol, dist} + B x_{tol, bot} \Rightarrow 150 \cdot 0.385 = 15 \cdot 0.007 + 135 x_{tol, bot}$$

$$\Rightarrow x_{tol, bot} = 0.427$$

Solving for mol frac of benzene & xylene:

$$\begin{cases} F z_{ben} = D x_{ben, dist} + B x_{ben, bot} \\ F z_{xy} = D x_{xy, dist} + B x_{xy, bot} \end{cases} \Rightarrow \begin{cases} 150 \cdot 0.92 = 15 x_{ben, dist} + 135 x_{ben, bot} \\ 150 \cdot 0.095 = 15 x_{xy, dist} + 135 x_{xy, bot} \end{cases}$$

$$\begin{cases} x_{xy, dist} + x_{ben, dist} + x_{tol, dist} = 1 \\ x_{xy, bot} + x_{ben, bot} + x_{tol, bot} = 1 \end{cases}$$

$$\begin{cases} x_{ben, dist} + x_{xy, dist} + 0.007 = 1 \\ x_{ben, bot} + x_{xy, bot} + 0.427 = 1 \end{cases}$$

$\Rightarrow$ Not enough independent eqns.

$$FR_{tol, bot} = \frac{B x_{tol, bot}}{F z_{tol}} = \frac{135 \cdot 0.427}{150 \cdot 0.385} = 0.998$$

$BP_{tol} < BP_{xylene} \Rightarrow$ Can assume toluene to be HK, xylene HNK and benzene HNK

$$\Rightarrow FR_{xy,bot} = 1 \Rightarrow \frac{B x_{xy,bot}}{F z_{xy}} = 1 \Rightarrow \frac{135 x_{xy,bot}}{150 \cdot 0.095} = 1 \Rightarrow x_{xy,bot} = 0.106$$

$$xylene\ HNK \Rightarrow x_{xy,dist} = 0$$

$$x_{ben,dist} = 1 - x_{tol,dist} - x_{xy,dist} = 1 - 0.007 - 0 = 0.993$$

$$x_{ben,bot} = 1 - x_{tol,bot} - x_{xy,bot} = 1 - 0.427 - 0.106 = 0.467$$

$$b) \alpha_{LK-HK} = \alpha_{ben-tol} = 2.22$$

$$\left(\frac{x_{LK}}{x_{HK}} \right)_{dist} = \alpha_{LK-HK}^{N_{min}} \left(\frac{x_{LK}}{x_{HK}} \right)_{bot} \Rightarrow \frac{0.993}{0.007} = 2.22^{N_{min}} \frac{0.467}{0.427}$$

$$N_{min} = 6.1 \approx 7 \text{ stages}$$

c) Reuse Fenske eqn for 1 sub HK with HNK:

$$N_{min} = \frac{\ln \left[\frac{(FR_{LK,dist})}{(1 - FR_{LK,dist})} \right] / \left(\frac{1 - FR_{HNK,bot}}{FR_{HNK,bot}} \right)}{\ln \alpha_{LK-HNK}}$$

$$FR_{LK,dist} = \frac{P x_{ben,dist}}{F z_{ben}} = \frac{15 \cdot 0.993}{150 \cdot 0.52} = 0.191$$

$$\Rightarrow 6.1 = \frac{\ln \left[\left(\frac{0.191}{1 - 0.191} \right) / \left(\frac{1 - FR_{HNK,bot}}{FR_{HNK,bot}} \right) \right]}{\ln (2.22 - 2.01)} \Rightarrow FR_{HNK,bot} = 0.99997$$

$$\Rightarrow 0.99997 = \frac{B x_{xy,bot}}{F z_{xy}} \Rightarrow 0.99997 = \frac{135 x_{xy,bot}}{150 \cdot 0.095} \Rightarrow x_{xy,bot} = 0.106$$

Make sense since $FR_{xy,bot} = 0.99997$, and our initial assumption was 1, these values are close enough, so it was reasonable that mol frac of xylene at bottom are identical.

6

$$d) F(1-q) = F_f = \sum_i \frac{\alpha_{i-ref} \cdot b_i}{\alpha_{i-ref} - \phi} = 150 \text{ kcal/l}$$

Choose xylenes as ref. compound

$$\alpha_{xy-xy} = 1, \quad \alpha_{tol-xy} = 2.01, \quad \alpha_{ben-xy} = \alpha_{ben-tol} \alpha_{tol-xy} = 2.22 \cdot 2.01 = 4.46$$

$$150 = \left(\frac{4.46 \cdot 0.92}{4.46 - \phi} + \frac{2.01 \cdot 0.0385}{2.01 - \phi} + \frac{1 \cdot 0.095}{1 - \phi} \right) 150$$

F.D.7

1.5

D

$$q = 1$$

$$F = 155 \text{ kmol/h}$$

$$FR_{4-dist} = 0.99$$

$$FR_{5-bot} = 0.983$$



$$\text{methane LNK } z_1 = 0.05$$

$$\text{ethane LNK } z_2 = 0.10$$

$$\text{n-butane LK } z_4 = 0.15$$

$$\text{n-pentane HK } z_5 = 0.22$$

$$\text{n-hexane ANK } z_6 = 0.22$$

$$\text{n-heptane HNK } z_7 = 0.26$$

All NK do not distribute

$$\Rightarrow FR_{1-dist} = FR_{2-dist} = FR_{6-bot} = FR_{7-bot} = 1$$

a) & b) All values and calculation are on the attached Excel file

$$T_{bubble F} = 114.8^\circ\text{C}$$

$$T_{bubble dist} = -78.05^\circ\text{C}$$

$$c) \frac{x_{4-dist}}{x_{5-dist}} = \frac{N_{min}}{\alpha_{4-5}} \frac{x_{4-bot}}{x_{5-bot}} \Rightarrow \frac{0.491}{0.012} = 4.10 \frac{N_{min}}{0.400}$$

$$\Rightarrow N_{min} = 6.13 \approx 7 \text{ stages}$$

$$d) F(1-q) = \sum_{i=1}^n \frac{\alpha_{i-5} F z_i}{\alpha_{i-5} - \phi} = 155 \text{ kmol/h}$$

$$d) F(1-q) = \sum_{i=1}^n \frac{\alpha_{i-5} F z_i}{\alpha_{i-5} - \phi} = 0 \Rightarrow \phi = 1.63 \text{ (since } \phi \in (1, 4.10) \text{ because ref = n-pentane)}$$

$$\Rightarrow V_{min} = \sum_i \frac{\alpha_{i-ref} D_{i-dist}}{\alpha_{i-ref} - \phi} = 60.93 \text{ kmol/h}$$

$$\Rightarrow L_{min} = V_{min} - D = 14.1 \text{ kmol/h} \Rightarrow \left(\frac{L}{D} \right)_{min} = 0.3$$

$$e) \frac{L}{D} = M \left(\frac{L}{D} \right)_{\min}$$

$$X = \frac{L/D - (L/D)_{\min}}{L/D + 1}$$

$$Y = \frac{1 - X^{0.0021}}{1 - 0.99357 X^{0.0021}}$$

$$N = (Y + N_{\min}) / (1 - Y)$$

Calculated steps in Excel

M	1.04	1.10	2.0
N	23	20	12

* Rounded up N values

f) Distillation such as this seems expensive and not economically viable, since the desired stream comes in at -14°C , it has to be cooled, then further cooled in the column up to -78°C which seems extremely expensive (I have no industrial experience?)

7.D.13

1.5

$$q=1$$

$$F = 100 \text{ kmol/h}$$

benzene LNK
toluene LK
cumene HK

$$z_b = 0.25$$

$$z_t = 0.3$$

$$z_c = 0.45$$



$$FR_{t, \text{dist}} = 0.99$$

$$\alpha_{B-T} = 2.5$$

$$\alpha_{C-T} = 0.21$$

$$FR_{c, \text{bot}} = 0.98$$

a) Assume $FR_{b, \text{dist}} = 1$

$$\Rightarrow D x_{b, \text{dist}} = F z_b = 100 \cdot 0.25 = 25 \text{ kmol/h}$$

$$FR_{t, \text{dist}} = 0.99 \Rightarrow D x_{t, \text{dist}} = 0.99 F z_t = 100 \cdot 0.3 \cdot 0.99 = 29.7 \text{ kmol/h}$$

$$FR_{c, \text{dist}} = 1 - 0.98 \cdot 0.02 \Rightarrow D x_{c, \text{dist}} = 0.02 \cdot 100 \cdot 0.45 = 0.9 \text{ kmol/h}$$

$$\Rightarrow D = D x_{b, \text{dist}} + D x_{t, \text{dist}} + D x_{c, \text{dist}} = 25 + 29.7 + 0.9 = 55.6 \text{ kmol/h}$$

$$x_{b, \text{dist}} = \frac{D x_{b, \text{dist}}}{D} = \frac{25}{55.6} = 0.450$$

$$x_{t, \text{dist}} = \frac{D x_{t, \text{dist}}}{D} = \frac{29.7}{55.6} = 0.534$$

$$x_{c, \text{dist}} = \frac{D x_{c, \text{dist}}}{D} = \frac{0.9}{55.6} = 0.016$$

$$F(1-q) = \sum \frac{\alpha_{i-t} F z_i}{\alpha_{i-t} - \phi} = 0 \Rightarrow \frac{2.5 \cdot 0.25}{2.5 - \phi} + \frac{1 \cdot 0.3}{1 - \phi} + \frac{0.21 \cdot 0.45}{0.21 - \phi} = 0$$

$$\Rightarrow \phi = 0.337 \quad (\text{choose } 0.21 < \phi < 1.5)$$

$$\Rightarrow V_{\text{min}} = \sum \frac{\alpha_{i-t} D x_{i, \text{dist}}}{\alpha_{i-t} - \phi} = \frac{2.5 \cdot 25}{2.5 - 0.337} + \frac{1 \cdot 29.7}{1 - 0.337} + \frac{0.21 \cdot 0.9}{0.21 - 0.337}$$

$$V_{\text{min}} = 72.23 \text{ kmol/h}$$

$$L_{\text{min}} = V_{\text{min}} - D = 16.63 \text{ kmol/h}$$

$$\Rightarrow (L/D)_{\text{min}} = 0.30$$

$$\alpha_{Lk-Hk} = \alpha_{t-c} = \frac{1}{\alpha_{c-t}} = \frac{1}{0.21} \approx 4.762$$

$$\frac{FR_{t,dot}}{FR_{H,dot}} = \alpha_{c-t}^{N_{min}} \frac{FR_{Lk,dot}}{FR_{Hk,dot}} \Rightarrow \frac{0.99}{1-0.98} = 4.762^{N_{min}} \frac{1-0.99}{0.98}$$

$$\Rightarrow N_{min} = 5.44 \text{ stages}$$

$$X = \frac{L/D - (L/D)_{min}}{L/D + 1}$$

$$Y = \frac{1 - X^{0.0031}}{1 - 0.99357 X^{0.0031}}$$

$$N(Y + N_{min}) / (1 - Y) = 9$$

Use Goalseek, we obtain $L/D = 0.90$

b) Using the same procedure as above we obtained L/D value nearly identical, N_{min} changed only slightly, not significant enough to have large changes on $(L/D)_{min}$ and as the result, L/D are identical, when $\alpha_{b-t} = 2.5$